

Name: _____

Student ID: _____

- ✓ Write down your student ID on the **top right corner** of each of the pages.
- ✓ Clearly write the solutions, along with the questions, on white paper with black ink (no need to use color pen, don't use pencils).
- ✓ Use **CamScanner**, or **Adobe Scan**, or **Microsoft Office Lens**, or any other software to scan the pages and make a **single PDF file**.
- ✓ After creating the PDF, make sure that (a) there are no pages missing, (b) all of the pages are legible, (c) your student ID on each page are visible.
- ✓ Please note, **collaboration ≠ copying**. You are allowed to discuss the questions and clear confusion you might have, but you have to write your solutions independently and be able to explain your answers during a random viva.
- ✓ **[Very Important]** Rename the PDF in the following format: "A1_StudentID_FullNameWithoutSpace.pdf". For example, if my student ID is 12345678 and my name is Shadman Shahid, the filename should be "A1_12345678_ShadmanShahid.pdf".
- ✓ **Submission Link:** <https://forms.gle/Ec4w1dmpmox9groVA>

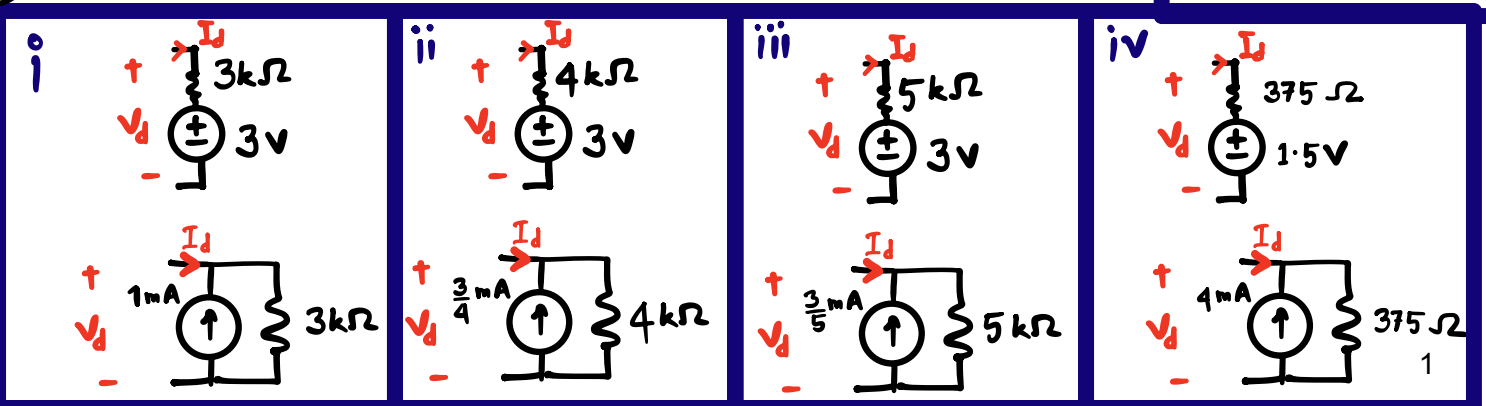
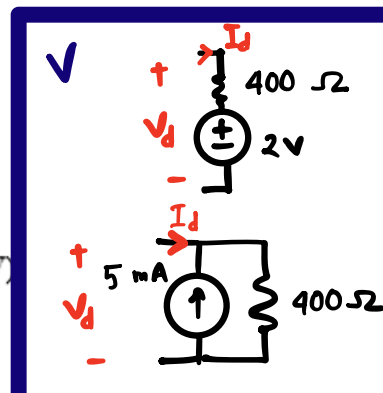
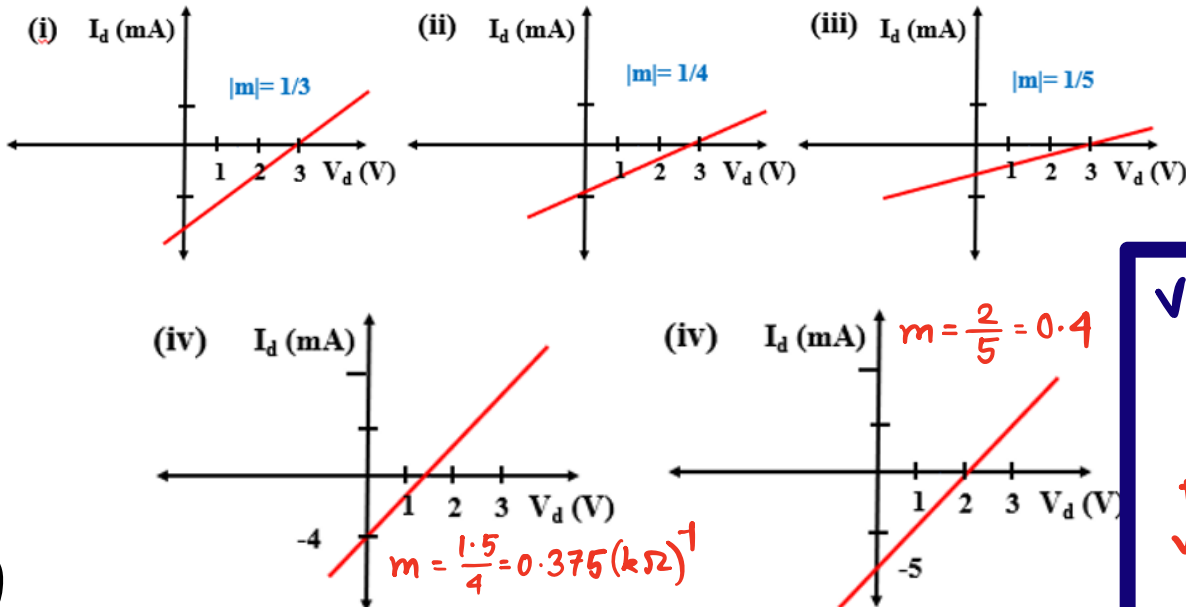
Question 1:

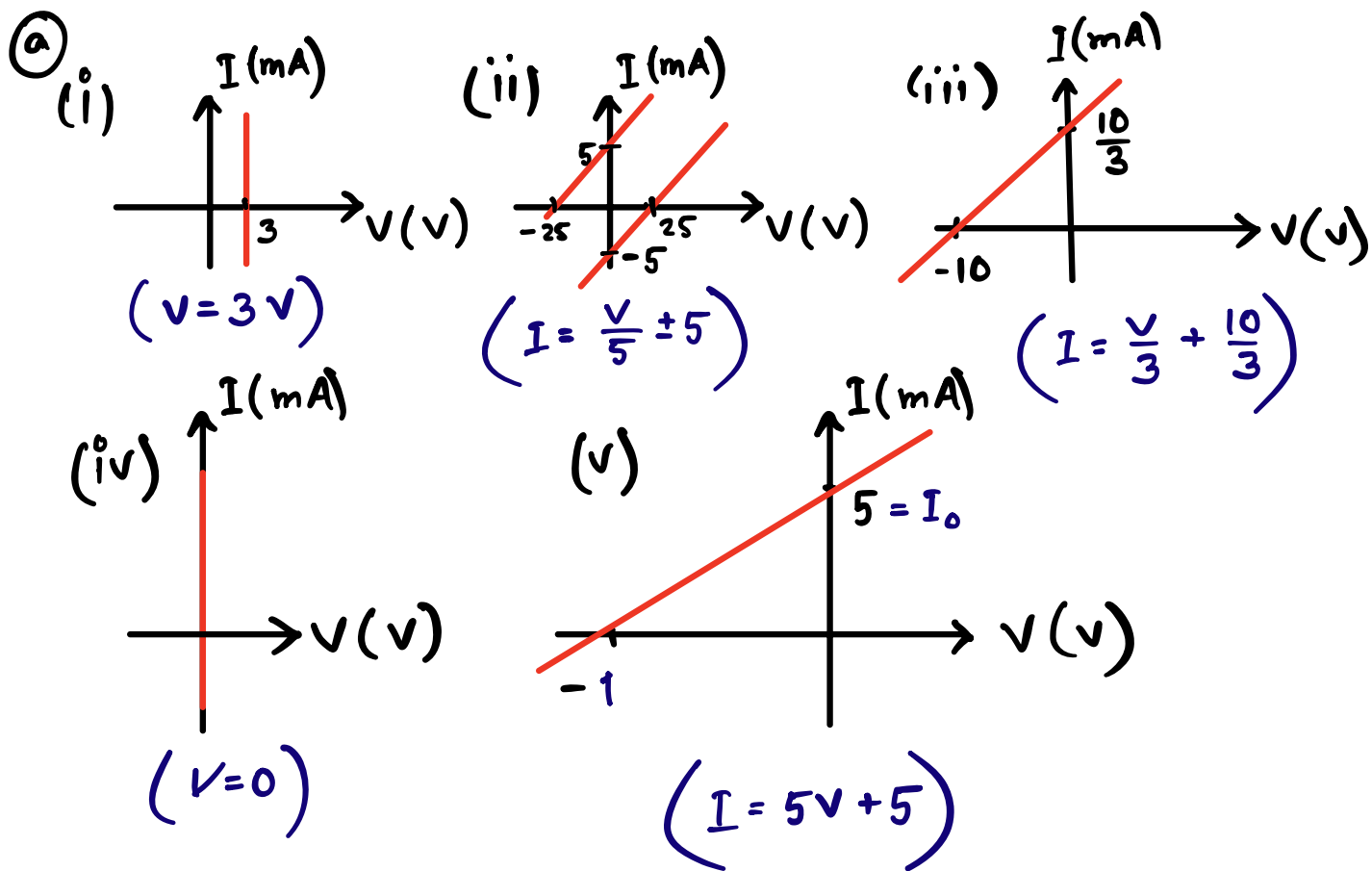
10 Marks

a) For the following sub-circuits, do the following: - Write down the equation representing this [CO1] 5
curve; - **Determine** the unknown parameters, - Label the I-V curve.

- A 3 V voltage source
- A 5 mA current source in parallel with a 5 k Ω
- A -10 V voltage source with 3 k Ω
- A short circuit
- A current Source, $I_o = 5$ mA in parallel with a resistor. The slope of the curve is,
 $m = 5 \text{ k}\Omega^{-1}$

b) **Draw** the sub-circuit that will result in the following IV characteristics. [CO3] 5





Question 2:

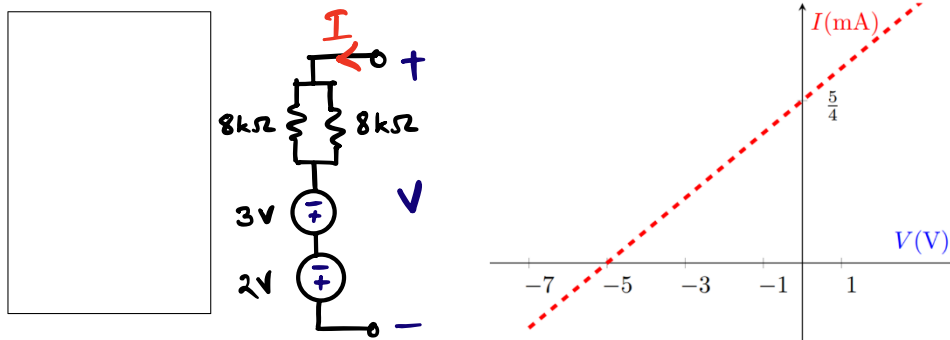
35 Marks

a) You are provided with the following circuit elements:

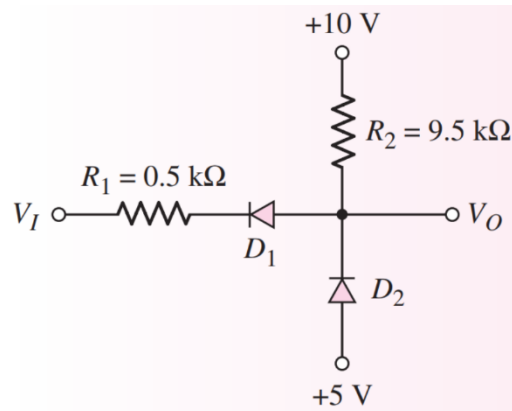
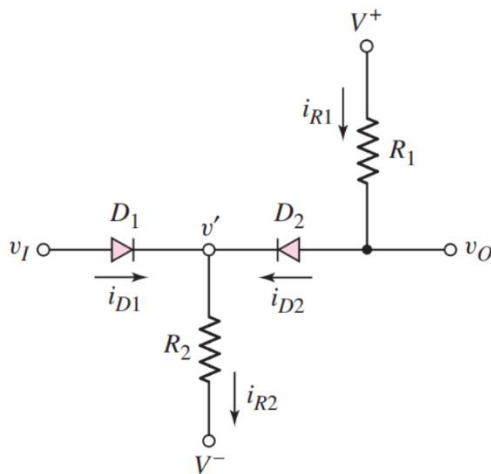
[CO3] 5

- Two $8\text{ k}\Omega$ resistors
- A 3 V voltage source
- A 2 V voltage source

Can you **implement** a circuit element **X** that has an IV-characteristics, as seen in the **right** figure below, but by **ONLY USING THE ELEMENTS MENTIONED ABOVE**? The voltage polarity and current direction should be as shown in the left figure.



b) For the circuit **below (left)**, assume the following circuit parameters: $R_1 = 5\text{ k}\Omega$, $R_2 = 10\text{ k}\Omega$, $V_{D0} = 0.7\text{ V}$, $V^+ = 5\text{ V}$, $V^- = -5\text{ V}$. Determine v_o , i_{D1} and i_{D2} for $v_I = 0$ and $v_I = 4\text{ V}$. [CO2] 3

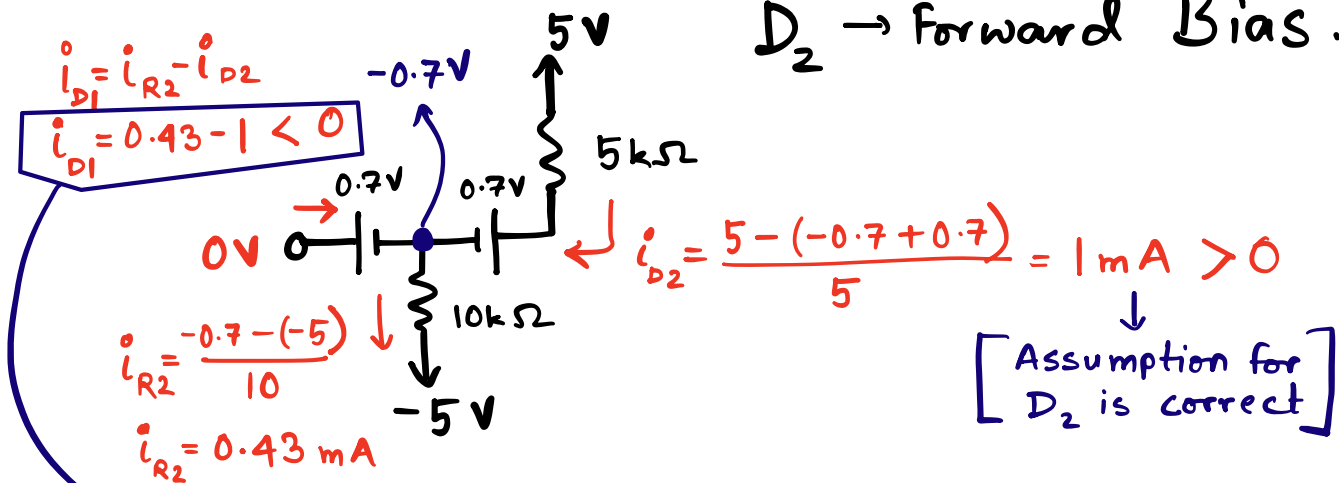


c) For the circuit **above (right)**., assume the following circuit parameters: $V_{D0} = 0.6\text{ V}$. Plot v_o versus v_I , for $0 \leq v_I \leq 10\text{ V}$. [CO2] 7

2(b) $\textcircled{1} \rightarrow V_I = 0V$

Initial assumption: $D_1 \rightarrow$ Forward Bias.

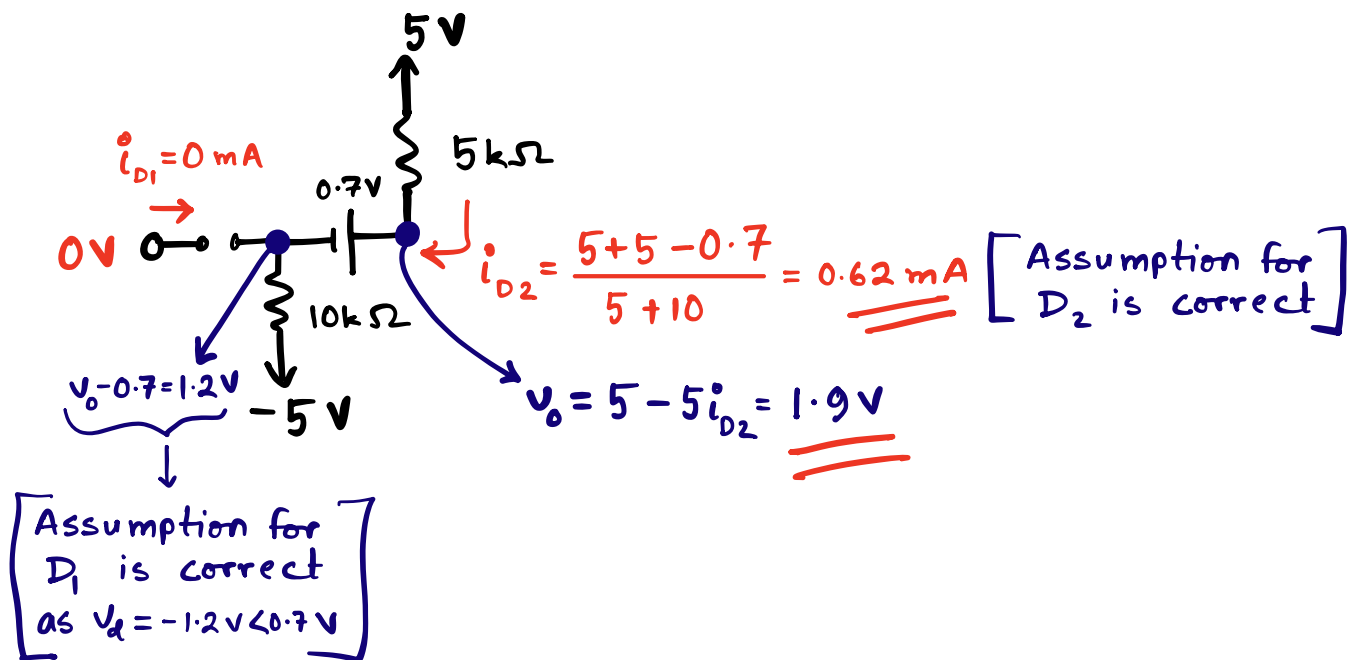
$D_2 \rightarrow$ Forward Bias.



$\rightarrow$ [Assumption for D_1 false] $\rightarrow$ New assumption:

$D_1 \rightarrow$ Reverse Bias.

$D_2 \rightarrow$ Forward Bias.



Question 2:

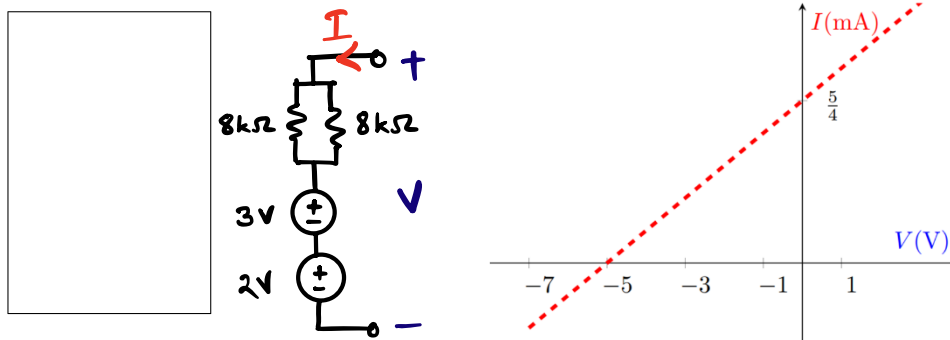
35 Marks

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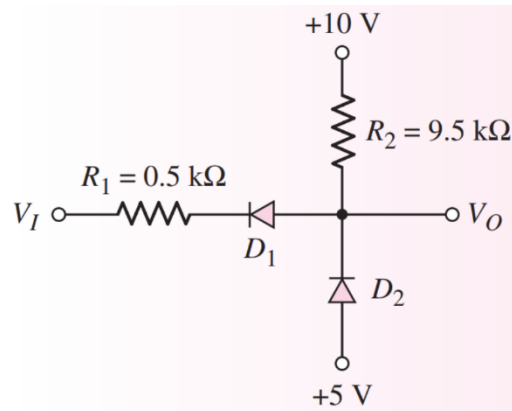
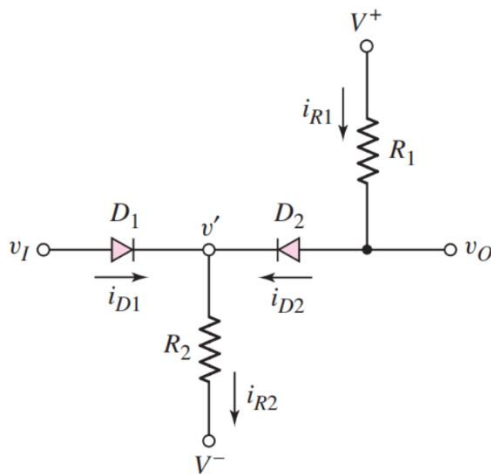
[CO3] 5

- Two $8\text{ k}\Omega$ resistors
- A 3 V voltage source
- A 2 V voltage source

Can you **implement** a circuit element **X** that has an IV-characteristics, as seen in the **right** figure below, but by **ONLY USING THE ELEMENTS MENTIONED ABOVE**? The voltage polarity and current direction should be as shown in the left figure.



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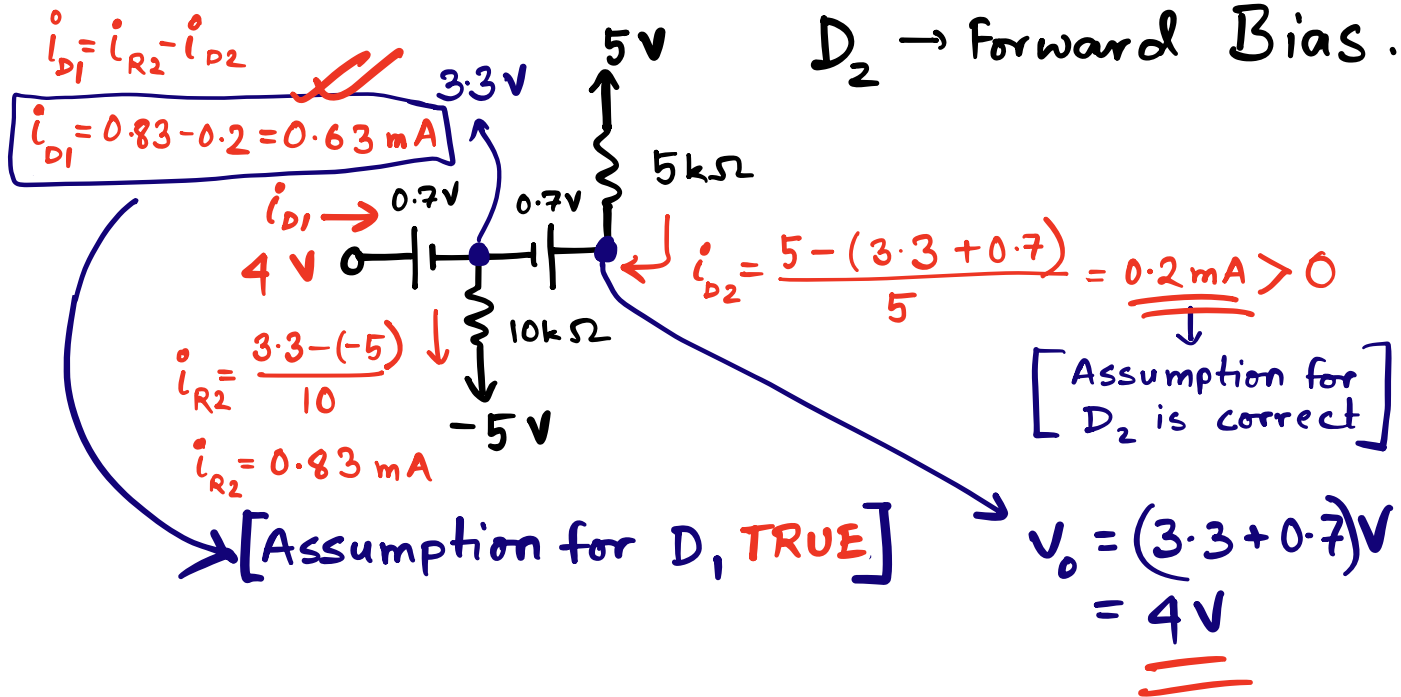


c) For the circuit **above (right)**., assume the following circuit parameters: $V_{D0} = 0.6\text{ V}$. Plot v_o versus v_I , for $0 \leq v_I \leq 10\text{ V}$. [CO2] 7

2(b) $\textcircled{1} \rightarrow V_I = 4 \text{ V}$

Initial assumption: $D_1 \rightarrow$ Forward Bias.

$D_2 \rightarrow$ Forward Bias.



2(c) Find out the graph in three stages.

Stage 1: D_1 : Forward biased.
 D_2 Forward biased.

Stage 2: D_1 : Forward biased.
 D_2 : Reverse biased.

Stage 3: D_1 : Reverse biased.
 D_2 : Reverse biased.

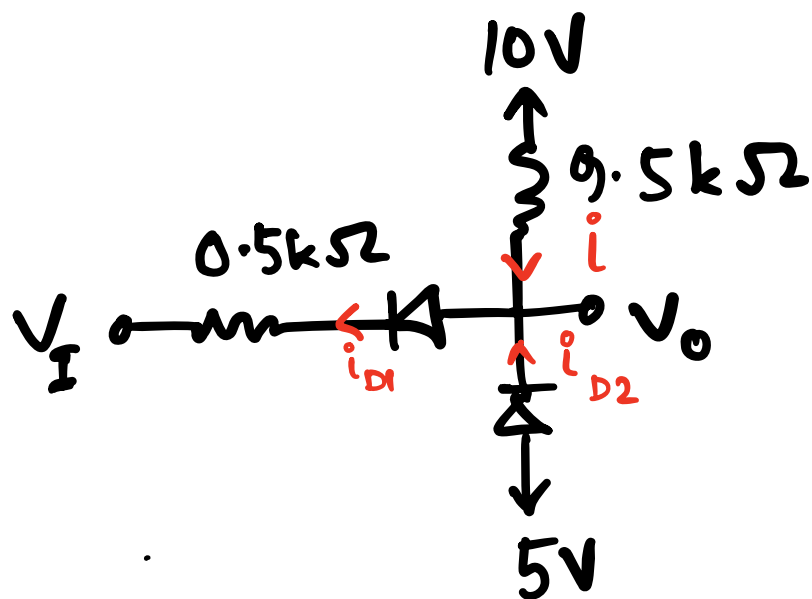
Stage 1

If D_1 is forward biased:

$$i_{D1} > 0$$

$$\Rightarrow \frac{V_o - V_{D0} - V_I}{0.5} > 0$$

$$V_o - V_I > 0.6 \text{ V}$$



If D_2 is forward biased,

$$V_o = 5 - V_{D0} = 4.4 \text{ V}$$

$$i_{D2} = i_{D1} - i > 0 \quad \rightarrow \quad i_{D1} > i$$

$$\Rightarrow \frac{V_o - V_{D0} - V_I}{0.5} - \frac{10 - V_o}{9.5} > 0 \quad \dots \dots (i)$$

At the edge of forward bias: $V_o = 4.4 \text{ V}$.

$$(i) \dots \Rightarrow \frac{4.4 - 0.6 - V_I}{0.5} > \frac{10 - 4.4}{9.5}$$

$$\Rightarrow \boxed{V_I < 3.505 \text{ V}}$$

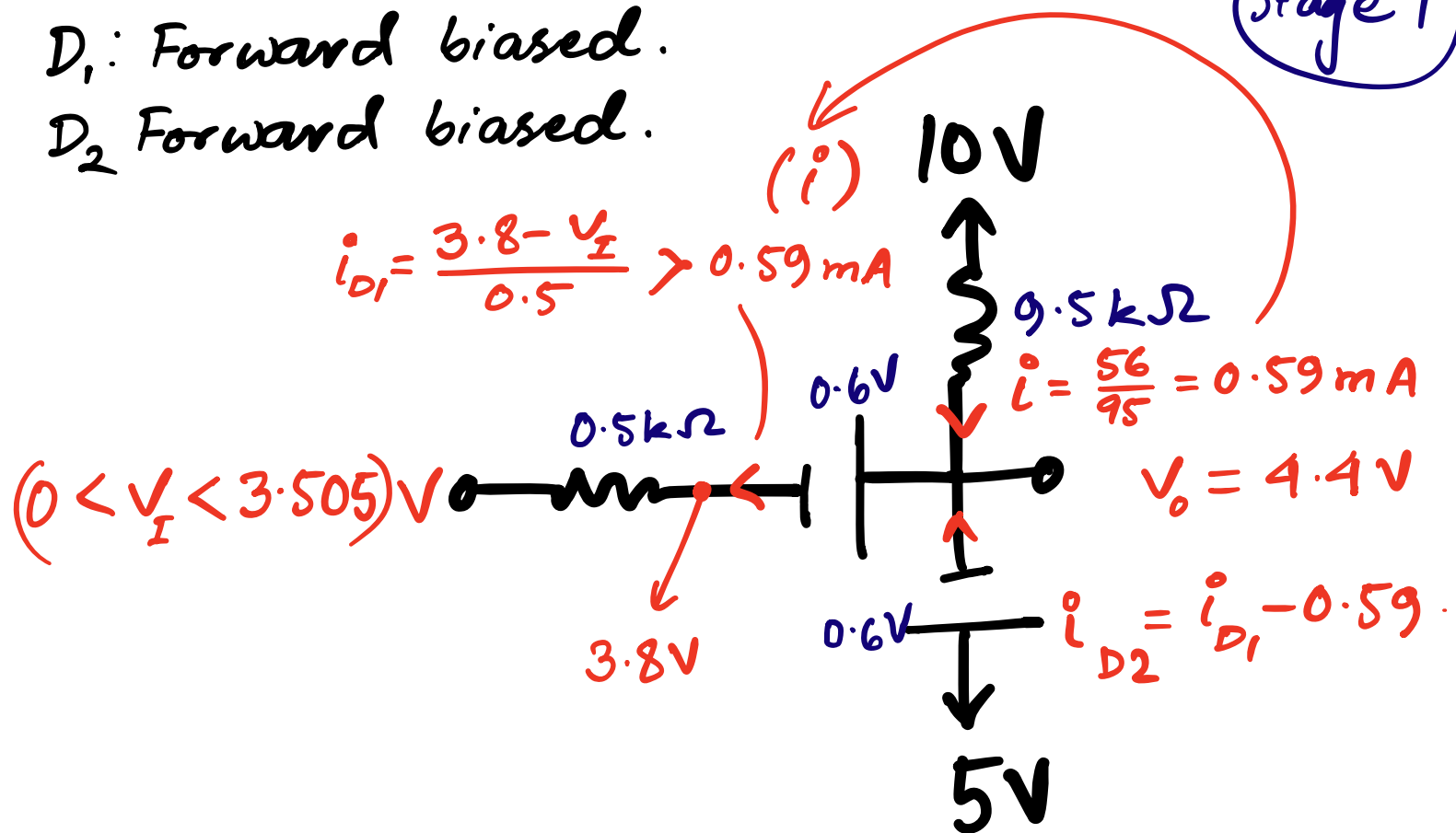
So, both D_1 & D_2 are f. B when:

$$V_I < 3.505 \text{ \& } V_o = 4.4 \text{ V \& } V_o - V_I > 0.6$$

D_1 : Forward biased.

D_2 Forward biased.

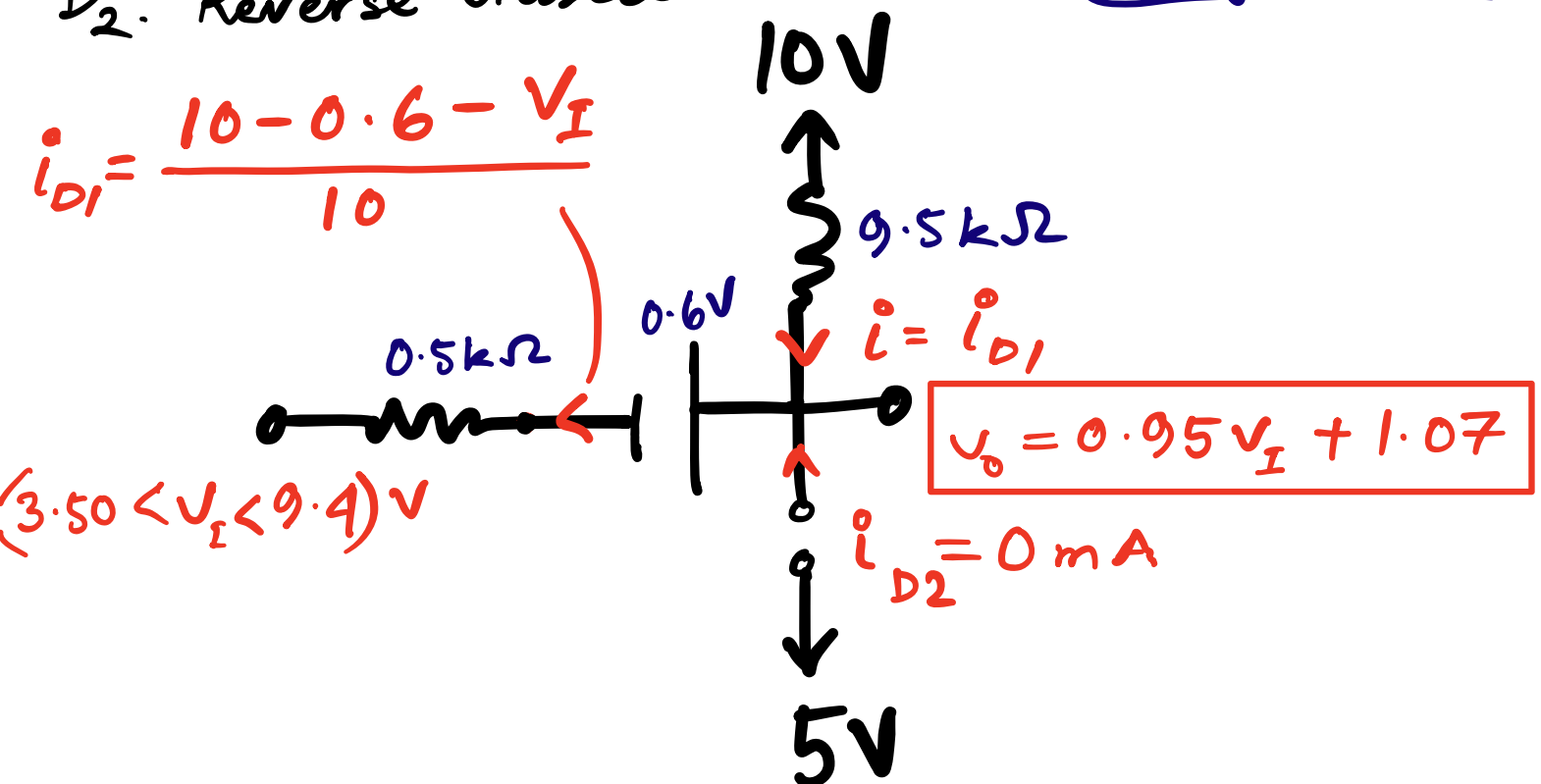
Stage 1



D_1 : Forward biased.

D_2 : Reverse biased.

Stage 2



Stage 2

When D_2 is Reverse biased: $V_I > 3.505 \text{ V}$

$i_{D_2} = 0$; $i_{D_1} = i$ (D_1 is still forward biased)

$$\Rightarrow \frac{V_o - 0.6 - V_I}{0.5} = \frac{10 - V_o}{9.5}$$

$$\Rightarrow \boxed{V_o = 0.95 V_I + 1.07}$$

Since D_1 is still forward biased.

$$V_o - V_I > 0.6$$

$$0.95 V_I + 1.07 - V_I > 0.6$$

$$\Rightarrow \boxed{V_I < 9.4 \text{ V}}$$

At the edge of forward biased D_1 : $V_I = 9.4 \text{ V}$

$$V_o = 0.95 V_I + 1.07 = 10 \text{ V}.$$

Stage 3:

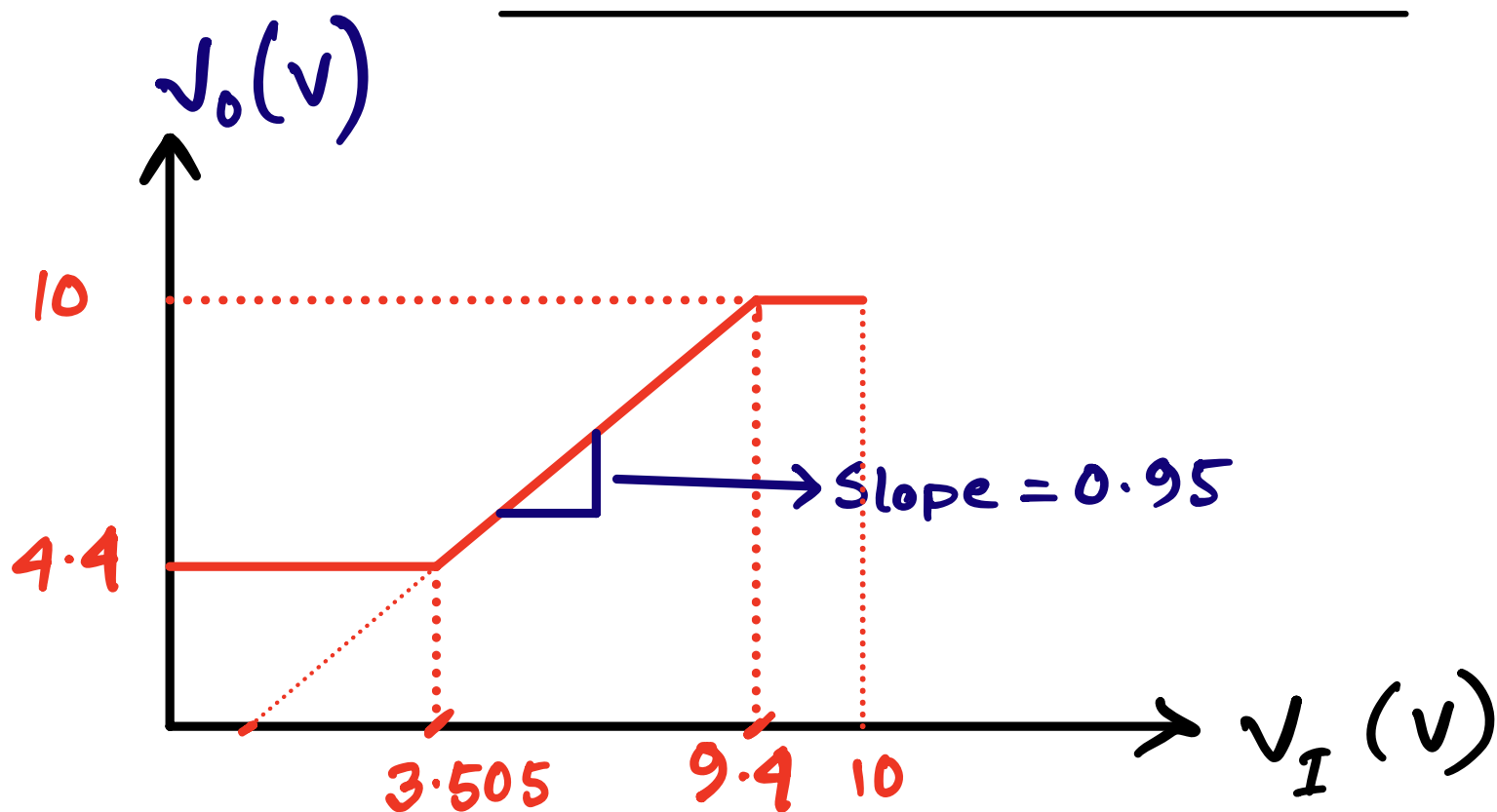
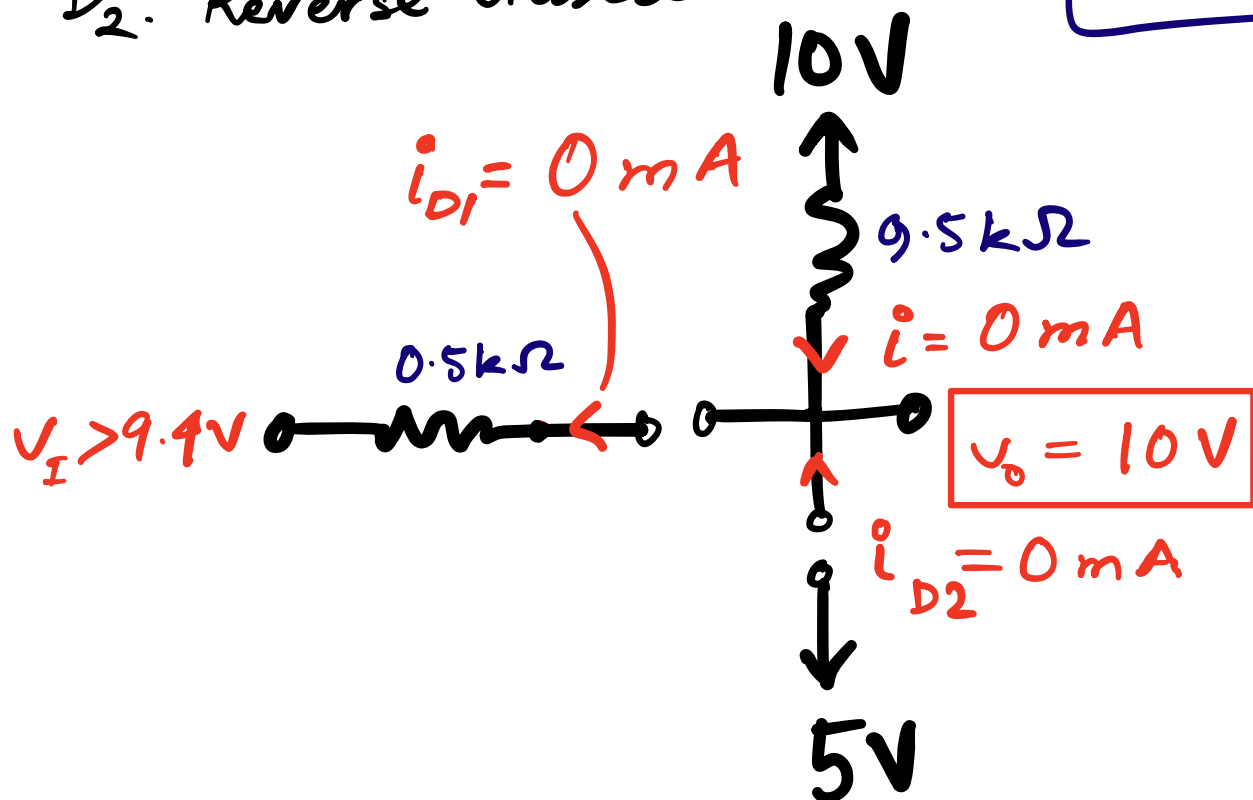
But if $V_o = 10 \text{ V}$; $i = 0$; $i_{D_1} = 0$; $i_{D_2} = 0 \rightarrow V_I > 9.4 \text{ V}$

Both D_1 & D_2 are Reverse biased.

D_1 : Reverse biased.

D_2 : Reverse biased.

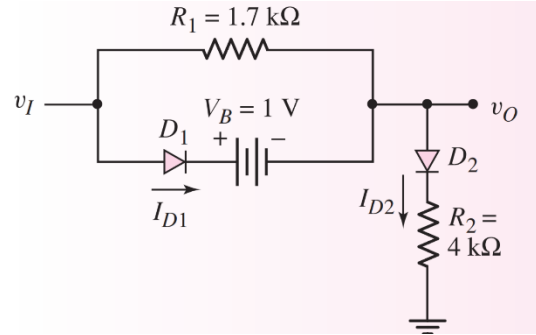
Stage 3



d) For the circuit below, the cut-in voltage for all the diodes are $V_{D0} = 0.7 \text{ V}$.

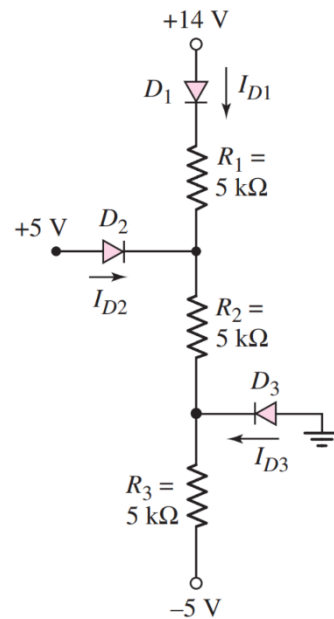
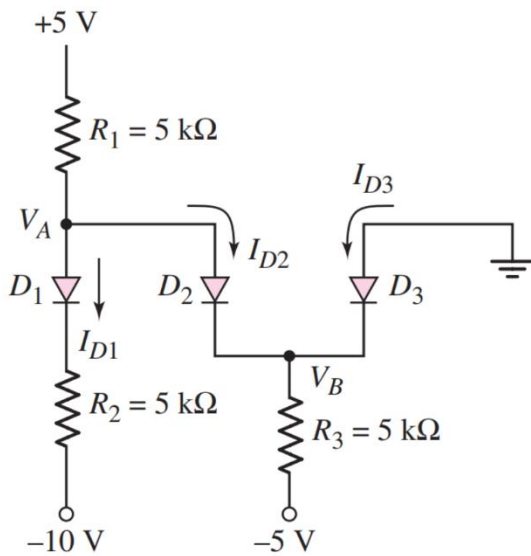
[CO2] 10

- Let $v_I = 5 \text{ V}$. Assume both diodes are conducting. Is this a correct assumption? Why or why not? Determine I_{R1} , I_{D1} , I_{D2} and v_O
- Repeat part (i) for $v_I = 10 \text{ V}$



e) Determine the current in each diode (I_{D1} , I_{D2} and I_{D3}) and the voltages V_A and V_B in the multi-diode circuit shown in the figure **below (left)**. Let, $V_{D0} = 0.7 \text{ V}$ for each diode.

[CO2] 5



f) Determine I_{D1} , I_{D2} and I_{D3} and the voltages V_A and V_B for the circuit **above (right)**. $V_{D0} = 0.7 \text{ V}$ for each diode.

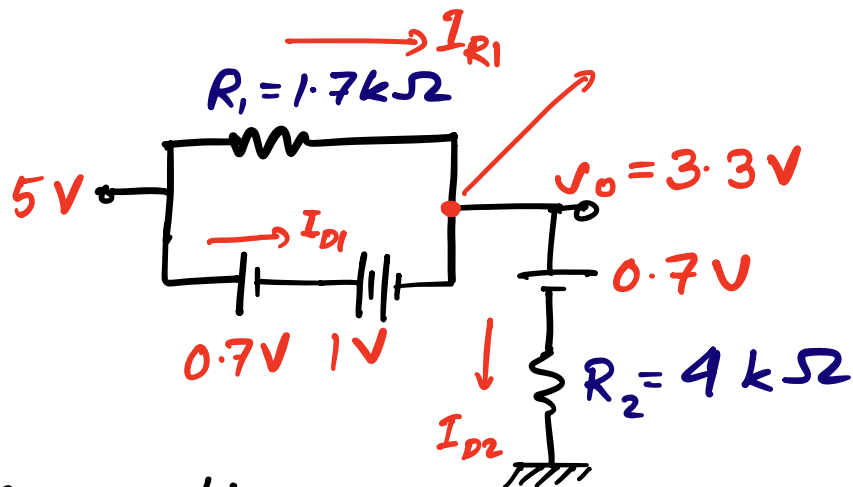
[CO2] 5

2(d) If both diodes are conducting:

$$V_o = (5 - 0.7 - 1) \text{ V} = 3.3 \text{ V}$$

$$I_{D2} = \frac{3.3 - 0.7}{4} \text{ mA}$$

$$= 0.65 \text{ mA} > 0$$



Assumption correct.

$$I_{D1} = I_{D2} - I_{R1}$$

$$= \left(0.65 - \frac{5 - 3.3}{1.7} \right) \text{ mA}$$

$$= 0.65 - 1 = -0.35 \text{ mA} < 0$$

Assumption false

Correct assumption: D_1 : Reverse biased.

D_2 : Forward biased

$$I_{D2} = I_{R1} = \frac{5 - 0.7}{4 + 1.7} \text{ mA}$$

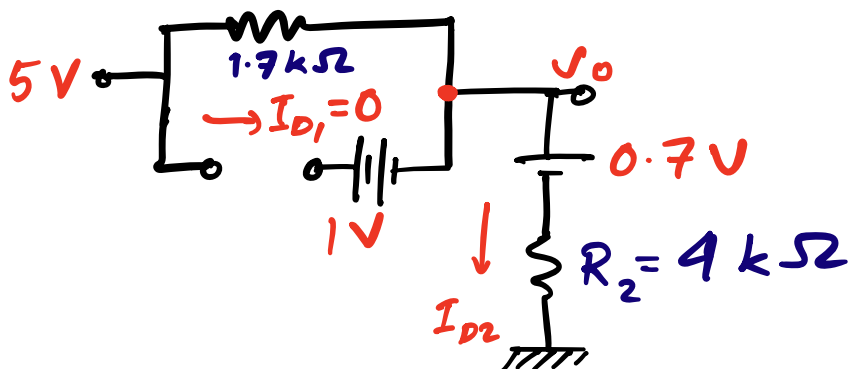
$$= 0.7544 \text{ mA}$$

$$V_o = (5 - 1.7 \times 0.7544) \text{ V}$$

$$= 3.717 \text{ V}$$

$$I_{D1} = 0 \text{ mA}$$

$$I_{R1} = I_{D2} = \frac{5 - 0.7}{5.7} = 0.75 \text{ mA}$$



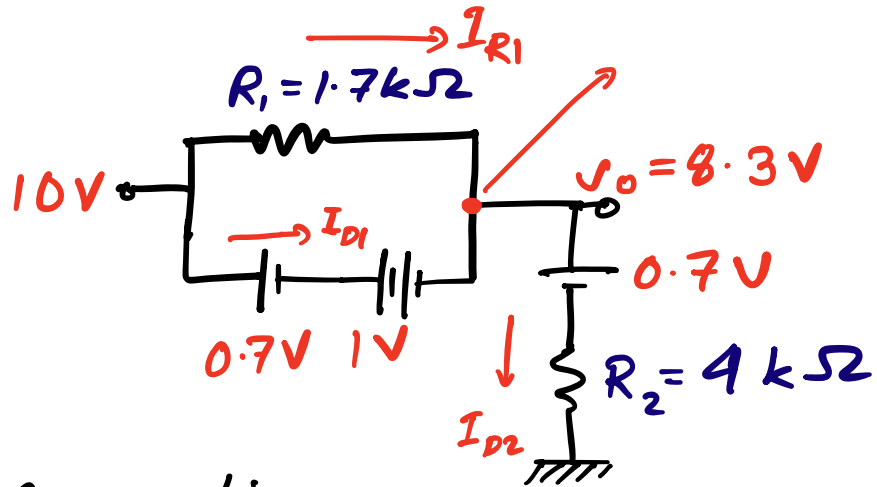
2(d)(ii)

If both diodes are conducting:

$$V_o = (10 - 0.7 - 1) \text{ V} = 8.3 \text{ V}$$

$$I_{D2} = \frac{8.3 - 0.7}{4} \text{ mA}$$

$$= 1.9 \text{ mA} > 0$$



Assumption correct.

$$I_{D1} = I_{D2} - I_{R1}$$

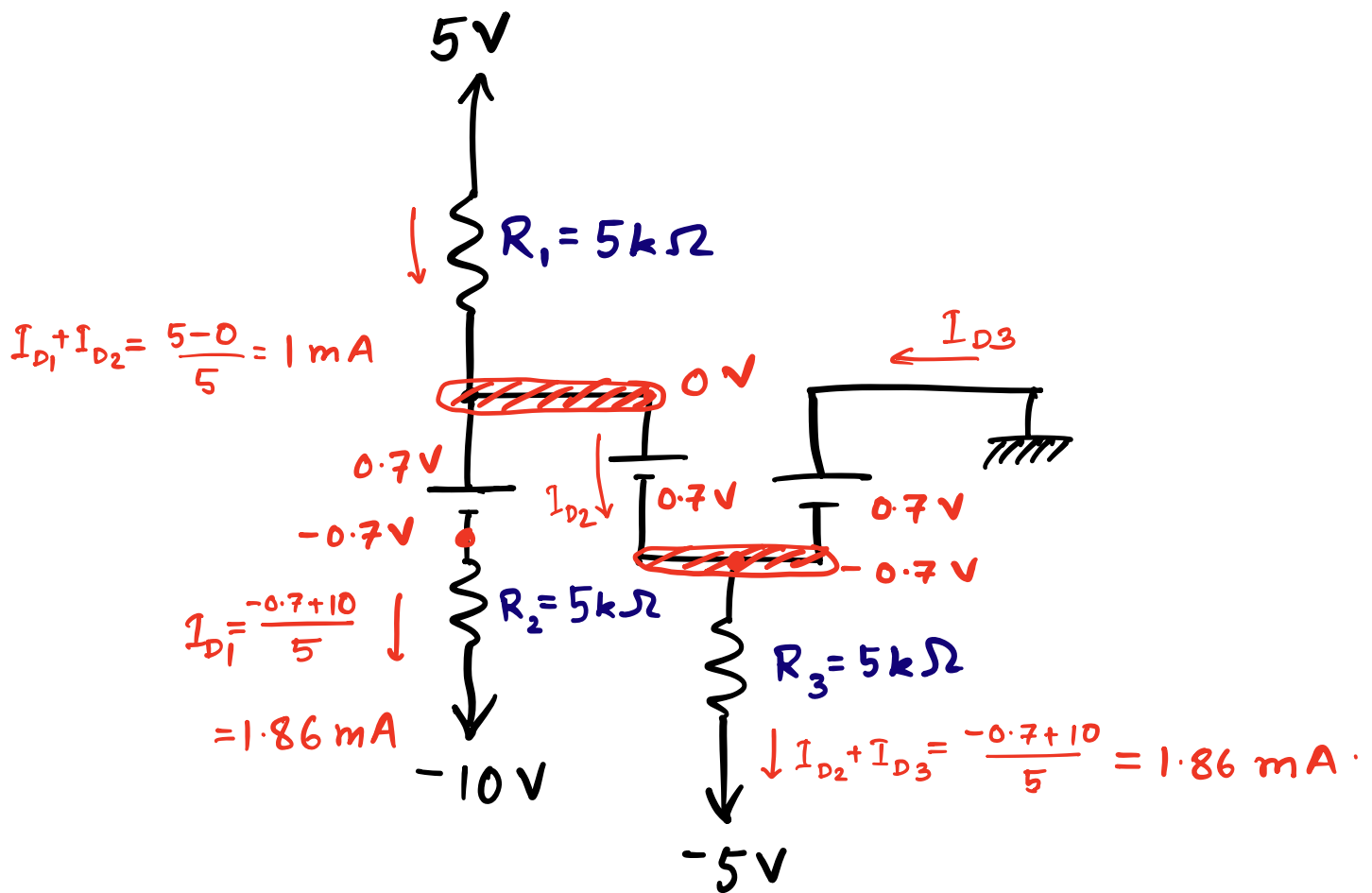
$$= \left(1.9 - \frac{10 - 8.3}{1.7} \right) \text{ mA}$$

$$I + I_{D1} = I_{D2}$$

$$= 1.9 - 1 = 0.9 \text{ mA} > 0$$

Assumption correct.

2(e) $D_1: F.B$
 $D_2: F.B$
 $D_3: F.B$



$$I_{D1} = 1.86 \text{ mA}$$

$$I_{D1} + I_{D2} = 1 \text{ mA}$$

$$\therefore I_{D2} = 1 - I_{D1} = 1 - 1.86 = -0.86 \text{ mA} \rightarrow \text{Assumption false}$$

$$I_{D3} + I_{D2} = 1.86 \text{ mA}$$

$$I_{D3} = 2.72 \text{ mA}$$

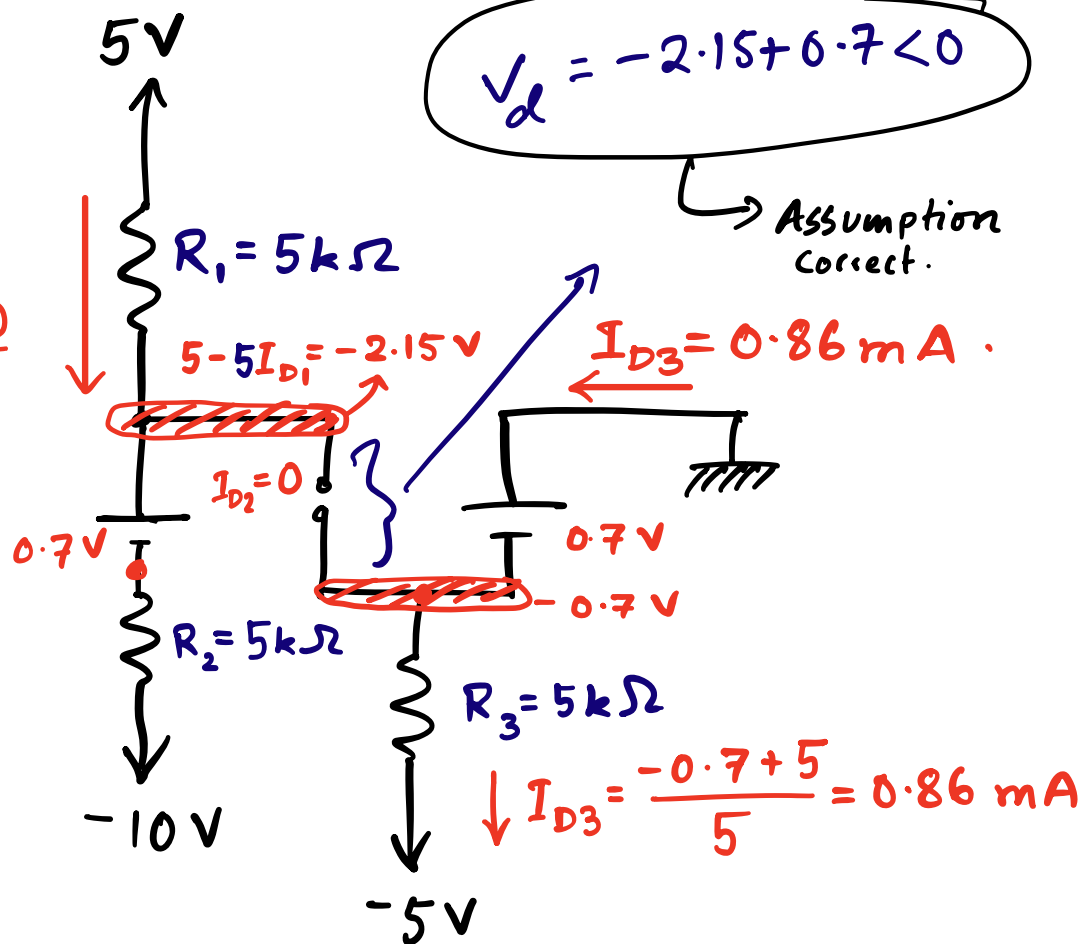
New Assumption:

$D_1: F.B$

$D_2: R.B$

$D_3: F.B$

$$I_{D1} = \frac{5 - 0.7 + 10}{5 + 5} \\ = 1.43 \text{ mA}$$



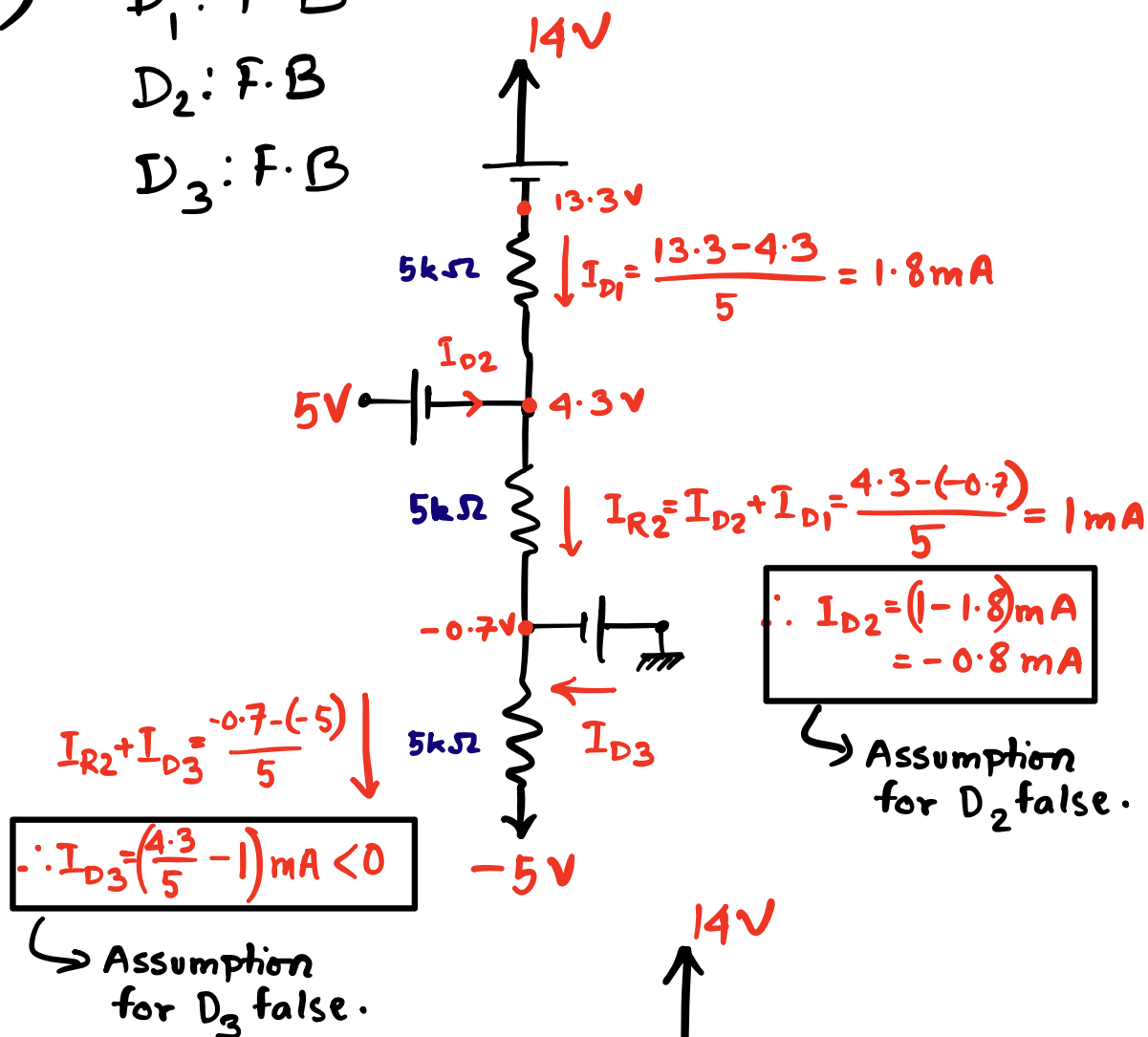
$$\left. \begin{array}{l} I_{D1} = 1.43 \text{ mA} \\ I_{D2} = 0 \\ I_{D3} = 0.86 \text{ mA} \end{array} \right\} \begin{array}{l} V_A = -2.15 \text{ V} \\ V_B = -0.7 \text{ V} \end{array}$$

2(f)

$D_1: F.B$

$D_2: F.B$

$D_3: F.B$

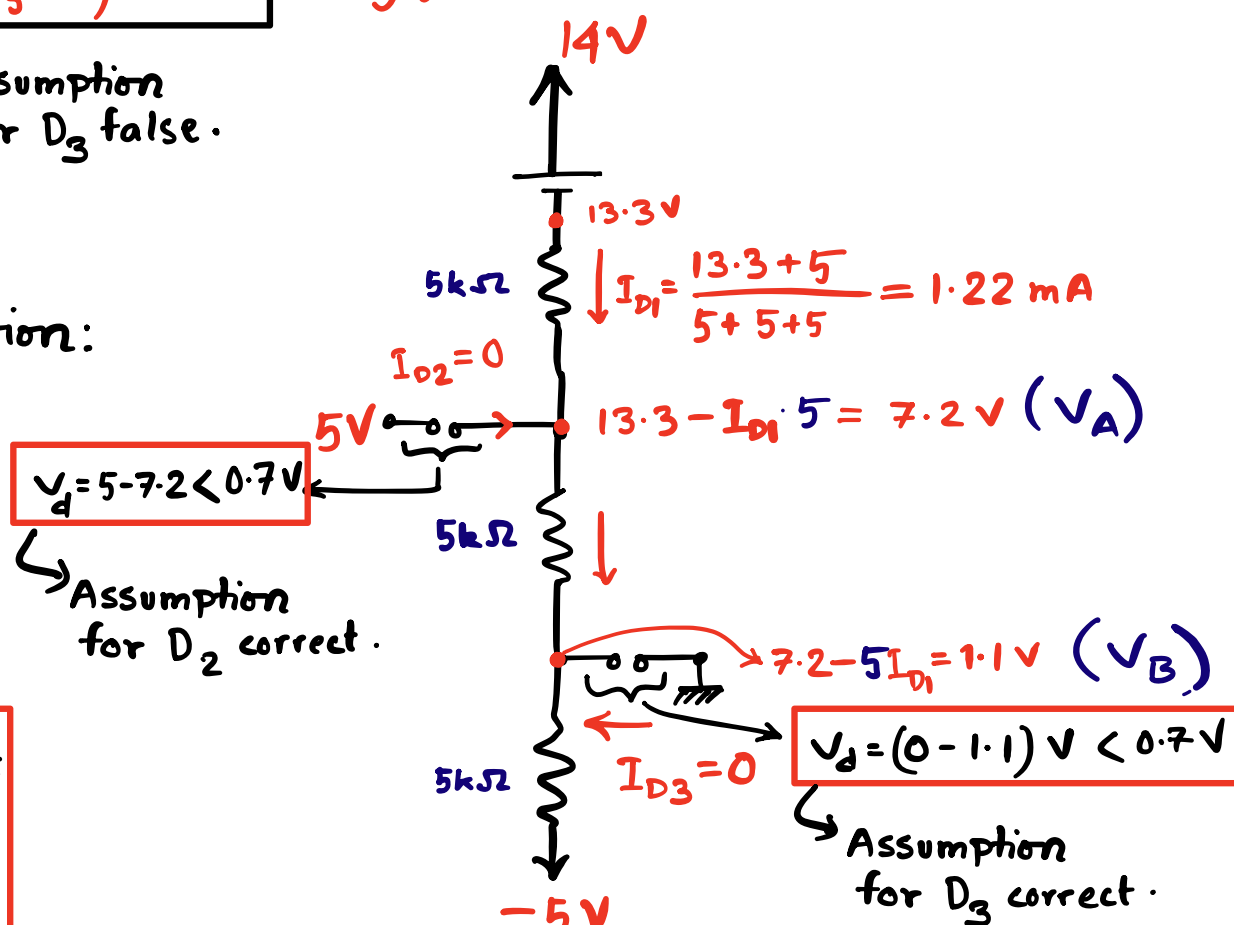


New Assumption:

$D_1: F.B$

$D_2: R.B$

$D_3: R.B$



$$I_{D1} = 1.22 \text{ mA}$$

$$I_{D2} = I_{D3} = 0$$

$$V_A = 7.2 \text{ V}$$

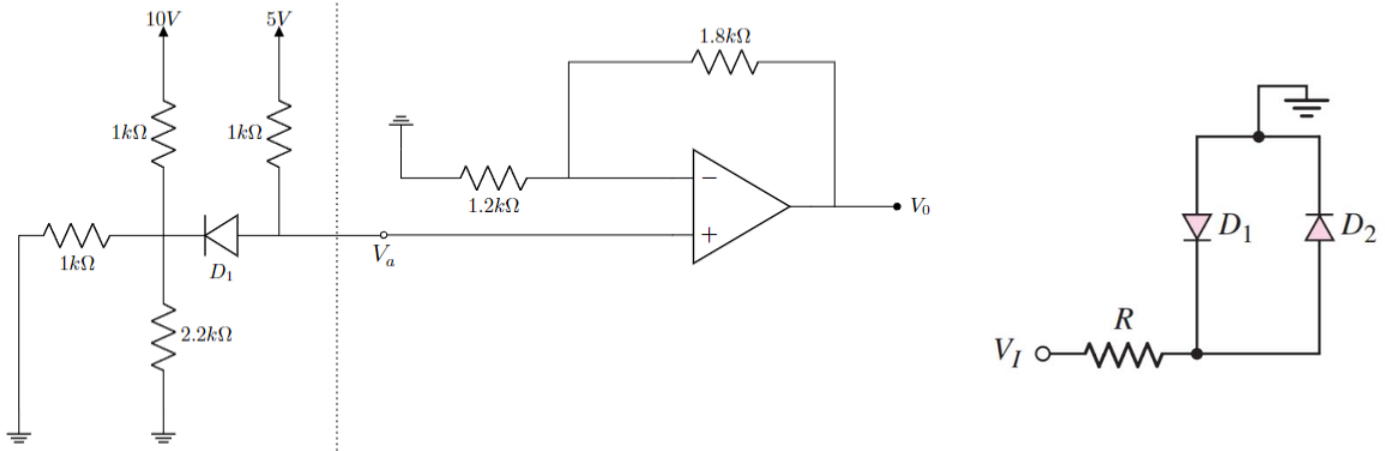
$$V_B = 1.1 \text{ V}$$

Question 3:

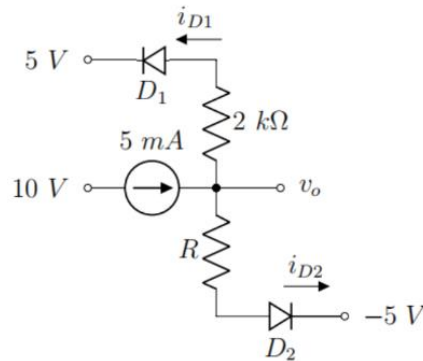
20 Marks

The saturation voltages of the Op-Amp on the **left below**, are given as $+V_{\text{sat}} = +10 \text{ V}$ and $-V_{\text{sat}} = -10 \text{ V}$. The forward voltage drop of the diode, V_{D0} is 0.7 V .

- Determine** the operating mode diode, D_1 . Verify your assumption with necessary calculations.
- Calculate** the voltage at –
 - Node ' V_a '
 - Non-inverting terminal of the Op-Amp,
 - Inverting terminal of the Op-Amp.
- Find out the output voltage, V_o of the Op-Amp.

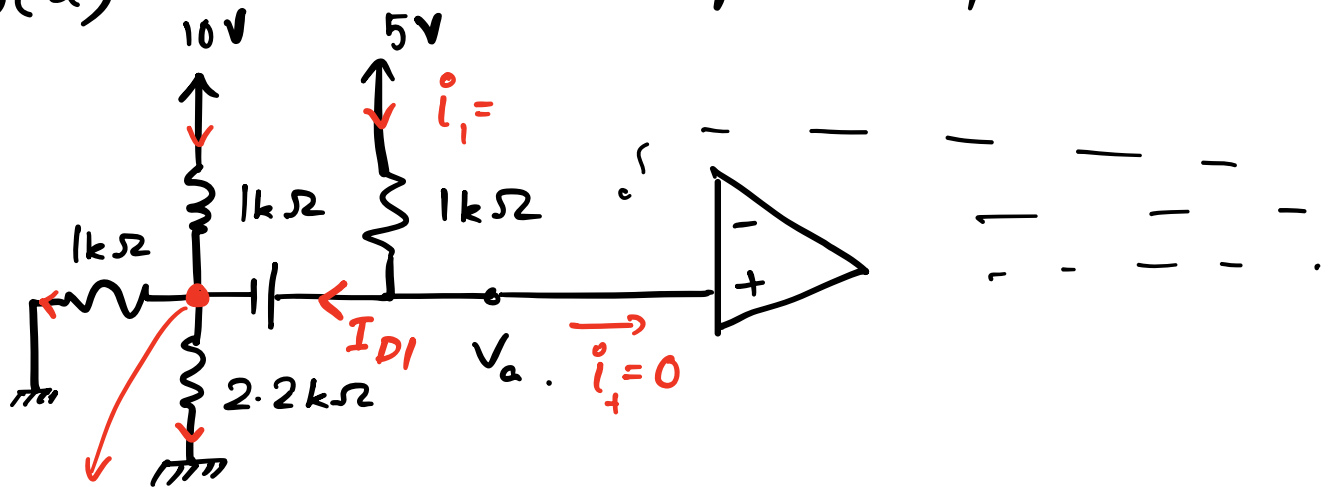


- The parameters of D_1 and D_2 in the circuit shown in **above right** are $V_{D0} = 1.7 \text{ V}$ and $r_f = 20 \Omega$. The current in each diode is to be limited to $I_D = 15 \text{ mA}$ for $V_I = \pm 5 \text{ V}$. **Determine** the required value of R . [CO3] 5
- Find V_o , i_{D1} and i_{D2} for $R = 1 \text{ k}\Omega$. Assume diode constant voltage drop model with $V_{D0} = 0.7 \text{ V}$. In each case, write down the states of the diodes (ON/OFF). You must verify your assumptions. [CO2] 5



3(a)

Assumption: D_1 is in F.B.



$$V_a - 0.7$$

Node Equation at Supernode $V_a (V_a - 0.7)$:

$$\frac{5 - V_a}{1} = \frac{V_a - 0.7}{1} + \frac{V_a - 0.7}{2.2} - \frac{10 - (V_a - 0.7)}{1}$$

$$V_a + (V_a - 0.7) \left(2 + \frac{1}{2.2} \right) = 5 + 10$$

$$V_a = \frac{15 + \frac{189}{110}}{\frac{38}{11}} \text{ V} = 4.839 \text{ V}$$

b (i)

$$I_{D1} = \frac{5 - 4.839}{1} > 0 \text{ mA}$$

→ Assumption about D_1 true.

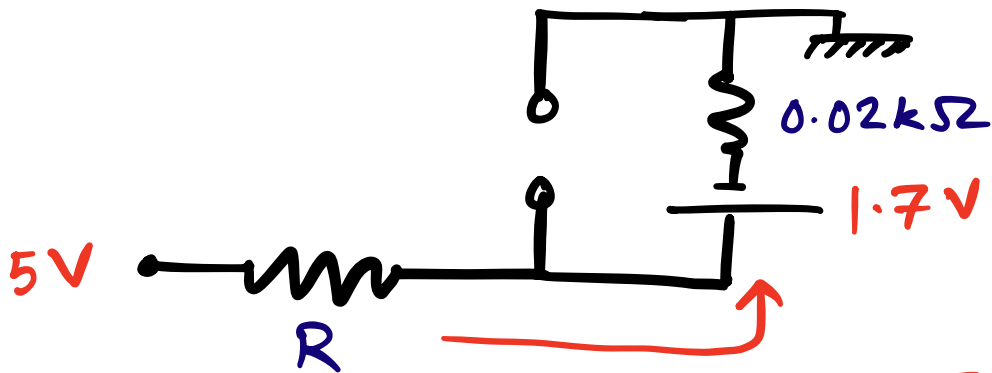
b (ii) $V_a = 4.839 \text{ V} = V_{(+)}$

(iii) $V_{(-)} = V_{(+)} = V_a = 4.839 \text{ V}$

c $V_o = \left(1 + \frac{1.8}{1.2} \right) V_a = 12.099 \text{ V}$

3 (d) From the configuration, it is obvious that Both diodes cannot be simultaneously in Forward bias.

When $V_I = 5V$: D_1 is R.B
 D_2 is F.B.



$$I_D = 15 \text{ mA} = \frac{5 - 1.7}{R + 0.02}$$

$$\Rightarrow \frac{3.3}{R + 0.02} = 15$$

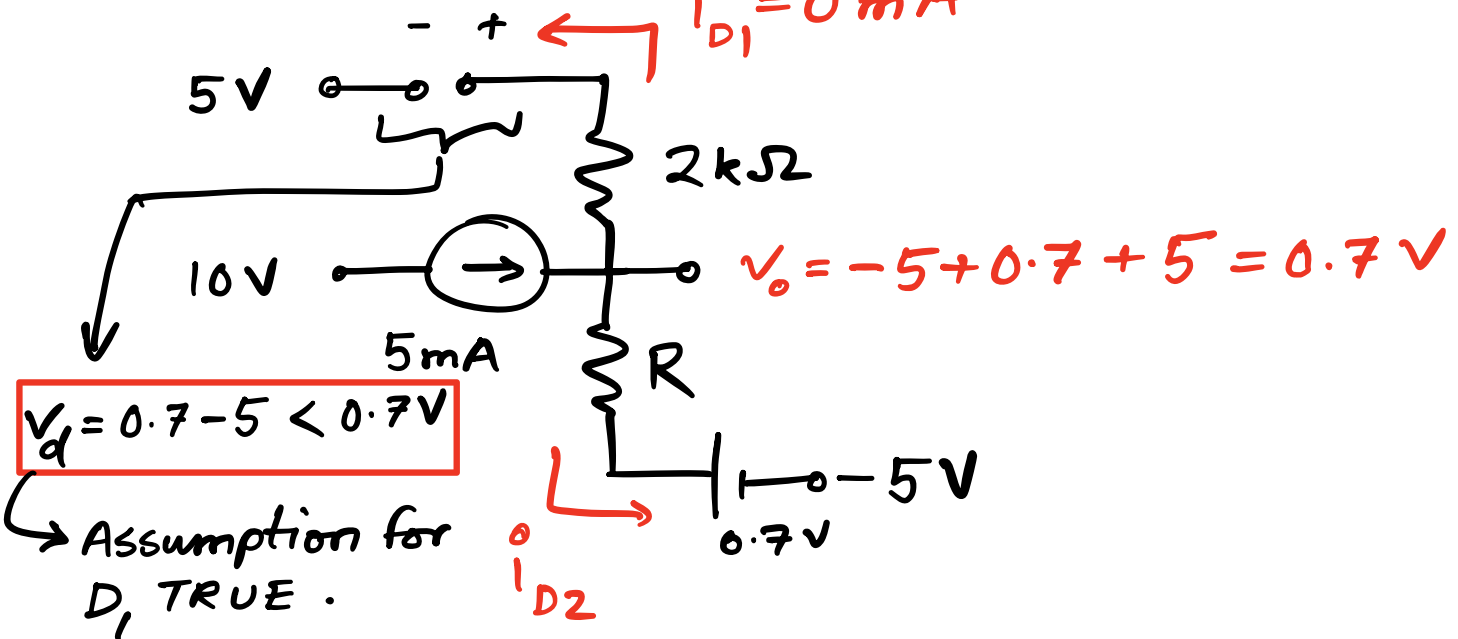
$$R = \frac{3.3 - 0.02 \times 15}{15} = 0.2 \text{ k}\Omega$$

$$\therefore R \geq 0.2 \text{ k}\Omega$$

$$\text{or, } R \geq 200 \Omega$$

3 (e) Assuming: $D_1: R.B$
 $D_2: F.B$

$$i_{D1} = 0 \text{ mA}$$



$$i_{D1} + i_{D2} = 5$$

$$i_{D2} = 5 \text{ mA}$$

$$\therefore V_o = 0.7 \text{ V}$$